

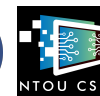
Economics and Computation

Introduction

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Textbooks and Materials

- Textbooks/Lectures:

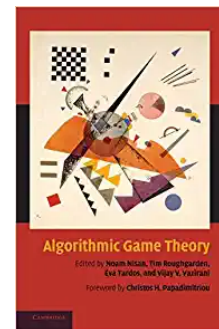
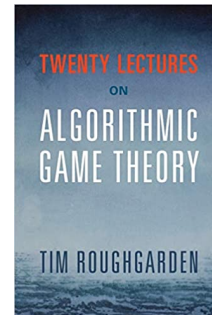
- *Twenty Lectures on Algorithmic Game Theory*. Tim Roughgarden. Cambridge University Press. 2016.
- *Algorithmic Game Theory*. Noam Nisan, Tim Roughgarden, Eva Tardos, and Vijay V. Vazirani. Cambridge University Press. 2011.
- *Handbook of Computational Social Choice*. F. Brandt, Vincent Conitzer, U. Endriss, Jérôme Lang, Ariel D. Procaccia. Cambridge University Press. 2016.

- Other materials:

- *Essentials of Game Theory: A Concise, Multidisciplinary Introduction* (Synthesis Lectures on Artificial Intelligence and Machine Learning). Leyton- Brown and Kevin, Shoham. Cambridge University Press. 2008.

- Course website:

- https://josephcclin.github.io/courses/ec_and_agt.htm



Prerequisites

- Basic CS undergraduate courses in
 - Algorithms
 - Calculus
 - Probability theory
 - Programming
- Motivation.
- Curiosity.

Major Topics

- Game Theory
- Equilibrium Concepts
- Computational Social Choice
- Mechanism Design
- Auctions
- * No-Regret Dynamics

Grading Policy

- Attendance (10%)
- Four quizzes on assignments (60%)
 - Assignments announced first, then a quiz will be held in one week.
- Final presentation (30%)
 - Research conference papers
 - About December 14 & 21, 2026.

Grading policy for the presentations

- 2~3 people in each group.
- **Complete the presentation:** 65 point
 - Duration for each presentation: 30~45 minutes.
 - 2-person group: 30-35 minutes.
 - 3-person group: 40-45 minutes.
- **Raising questions:** +2 point for each one, maximum +10 point.
 - Check the count at the end of the class.
- **Responses to instructor questions:** +5 point for each one, maximum +15 point.
 - One individual will be designated to answer.
- **(Intra-Group) Peer Assessment:** maximum +10 point.

Source of the papers

- **International conferences:**
 - **EC**: Proceedings of the ACM conference on Electronic commerce.
 - **NeurIPS**: Conference on Neural Information Processing Systems.
 - **AAAI**: Association for the Advancement of Artificial Intelligence.
 - **IJCAI**: International Joint Conferences on Artificial Intelligence.
 - **AAMAS**: International Conference on Autonomous Agents and Multiagent Systems.
 - **SAGT**: Symposium on Algorithmic Game Theory.
 - **WINE**: International Conference on Web and Internet Economics.
 - Workshops such as **GAIW**, **CFD**, ...